

Waveserver Ai Installation: 4-Post 23-Inch Deep Equipment Frame

This quick installation sheet provides the procedure to install an AC or DC Waveserver Ai on a 4-post 23-inch-wide, deep equipment frame.

Safety



Danger

Risk of electrical shock

Before connecting the chassis to a power source, read all installation instructions.

Do not work on the unit or connect or disconnect cables during periods of lightning activity.

Before installing the unit, ensure that wiring at the installation area meets all local and national electrical codes.

All electrical interfaces on the unit are intended for intra building connections only. The interfaces are designed as type 2 and type 4 ports, as described in GR-1089-CORE, Issue 6, and require isolation from exposed outside plant cabling.



Danger

Risk of electrical shock

Ensure that the input power cable is disconnected from the power module before removal. Removing the power module while it is connected to live power may result in personal injury.



Danger

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



Danger

Risk of personal injury

Do not stack anything on top of the unit. The mounting brackets are designed to support only the weight of unit. Bracket failure due to excess weight can result in bodily injury and equipment damage.



Warning

Hazardous moving parts

Fan and power modules contain hazardous moving parts. Keep fingers and other body parts away.



Warning

Risk of burn

Take care when removing or replacing Waveserver Ai modules. The heat sinks of the modules are exposed to touch and are very hot.



Caution

Risk of damage to the chassis, modules, and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



Caution

Risk of equipment damage

Ensure that the input power cable is disconnected from the power module before removal. Removing the power module while it is connected to live power may result in damage to the module.

Required tools and supplies

- Waveserver Ai w/AC or DC power modules, fan modules
- 23-inch Bracket Kit, Deep Frame (186-1305-900)
- Short-handled #1 Phillips screwdriver
- Long-handled #2 Phillips screwdriver
- 1/4-in Phillips screwdriver (DC Waveserver Ai installations)
- Hex driver
- Rack screws
- Wire cutter and stripper
- Two (2) AC or two (2) DC power cables
- 6 AWG (16 mm²) wire for the chassis ground
- Crimp tool
- For DC Waveserver Ai installations, Ciena recommends using Molex 19164-0889 (ring lug with crimp terminal) to connect the DC power plant to the DC power module (186-1511-900)

Note: Use of the 240Vdc input on the AC/240Vdc power module (186-1510-901) is limited to China only.

Installation prerequisites

- The installation site meets all safety and electrical requirements.

Supported upright depths

The 186-1305-900 bracket kit can be installed in a 23-in EIA equipment frame. The following upright spacing depth is supported:

- 914 mm - 1067 mm (36-in - 42-in) (with and without ground bracket)

For more information about supported upright depths, see "Ordering" in the *Waveserver Ai User Guide*, 323-4002-100.

Unpacking Waveserver Ai equipment

1. Visually inspect the shipping container. If it is damaged, inform the shipping company and Ciena immediately.
2. Carefully remove the equipment from the shipping container and protective packaging, and place it on a flat, stable surface.
3. Visually inspect all equipment to ensure that all components have been included and are undamaged. If any component is damaged, inform the shipping company and Ciena immediately.
4. Waveserver Ai w/power modules, fan modules components are:
 - 1 x Waveserver Ai

- 2 x Waveserver Ai 100-240 Vac or AC/240Vdc or -48 Vdc Power Module (inserted at rear of chassis)
- 2 x Waveserver Ai Fan Module (inserted at rear of chassis)
- 1 x Waveserver Ai Control Subsystem (WCS) (inserted at rear of chassis)
- 1 x Waveserver Ai access panel (installed at front of chassis)
- Waveserver Ai Miscellaneous Hardware Kit
- 2 x Fiber Management Clip (packaged with mounting screws)

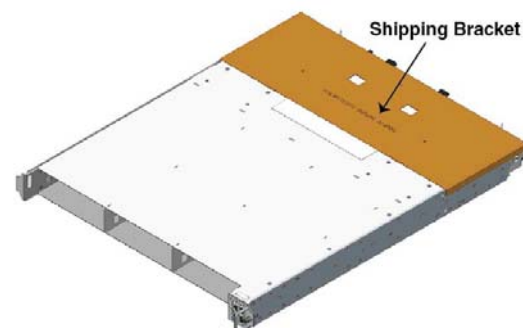
One or more of the following Waveserver Ai modules, depending on the pick-to-order kit purchased:

- Waveserver Ai 2x400G 56GBaud C-Band module (186-1600-900) (available in pick-to-order kits K86-3002-900 and K86-3003-900)
- Waveserver Ai 2x400G 56GBaud L-Band module (186-1800-900) (available in pick-to-order kit K86-3006-900)
- Waveserver Ai 2x400G 35/56GBaud C-Band module (186-1601-900) (available in pick-to-order kits K86-3100-900 and K86-3101-900)
- Waveserver Ai 2x400G 35/56GBaud L-Band module (186-1801-900) (available in pick-to-order kits K86-3106-900 and K86-3107-900)
- Waveserver Ai 1x400G 35/56GBaud C-Band module (186-1603-900) (available in pick-to-order kit K86-3102-900)

5. Note the following:

- A shipping bracket is attached to the rear of the chassis.

Figure 1: Shipping bracket at rear of chassis



- Dust covers are installed on the optical ports on the chassis. Do not remove these covers during installation.



Caution: Risk of damage to equipment

The power modules, WCS, and fan modules are not secured in their slots in the chassis. Ensure that the Waveserver Ai is placed on a flat, stable surface before removing the shipping bracket.

6. Using a #1 Phillips screwdriver, remove the seven (7) fasteners that secure the shipping bracket to the chassis, and carefully pull bracket away from the chassis.
7. Carefully remove the fan modules, Waveserver Ai Control Subsystem (WCS), and power modules from the chassis, and place them on a flat, stable surface.

Contents of Waveserver Ai 23-inch Bracket Kit, Deep Frame (186-1305-900)

Rear Cable Management Bar (410-6684-001) x 1



Front Bracket Guide (410-7667-001) x 2



Rear Bracket Guide (410-7668-001) x 2



T-Bar (410-6760-001) x 2



Alignment Bracket (410-7782-001) x 1



Mounting Bracket (410-7774-001) x 2



Right and Left Mounting Brackets (410-7773-001 and 410-7773-002) (not for Waveserver Ai installations)



Ground Bracket (410-6903-002) x 1

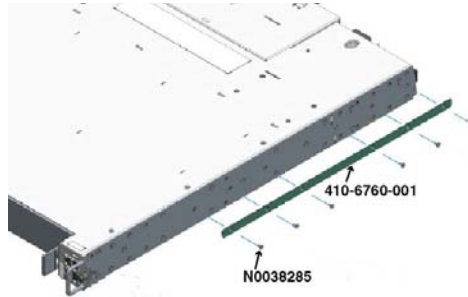


Fasteners

**Installation steps**

1. Install the T-Bars (410-6760-001) on both sides of the Waveserver Ai chassis.

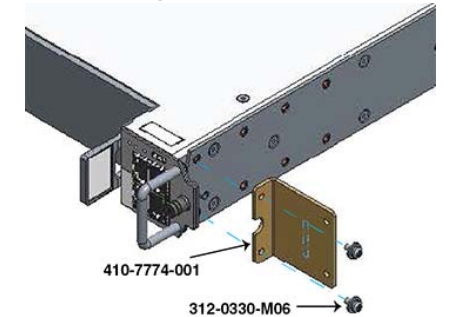
Position the T-Bar against the side of the chassis so that the pilot holes on the T-Bar align with the pilot holes on the chassis, and secure the T-Bar using six (6) countersink fasteners (N0045005). Repeat on the other side of the chassis.

Figure 2: T-Bar installation

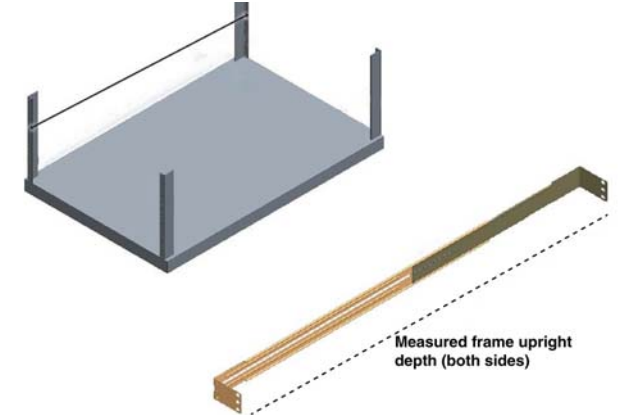
2. Install the frame mounting brackets (410-7774-001) on the front of the Waveserver Ai chassis.

Position a front frame mounting bracket at the front of the chassis so that the narrow edge of the tab with the two pilot holes is against the chassis and the larger tab is facing the back of the chassis. Secure the bracket using two (2) Phillips pan head screws (312-0330-M06). Repeat on the other side of the chassis.

Note: The flush-mount setback option is supported, with rail assemblies adjustable to 36.4" - 42.4".

Figure 3: Frame mounting bracket installation

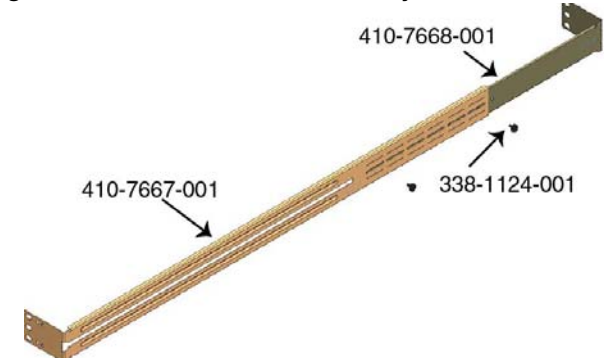
3. Measure the rack upright mounting surface spacing distance (front to back) on both sides of the equipment frame.

Figure 4: Measured frame upright depth

4. Assemble the front (410-7667-001) and rear (410-7668-001) bracket guides.

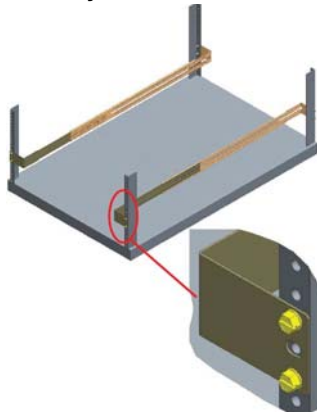
Insert the end of a rear bracket guide into the end of a front bracket guide. Adjust the length so that the distance between the front flange and rear flange is approximately the same as the distance measured in the previous step.

Loosely fasten using two (2) plastic head thumb screws (338-1124-001). The thumb screws should be placed as far apart as possible to keep the assembly square.

Figure 5: Front and rear rail assembly

5. Install the rail assemblies on the equipment frame.

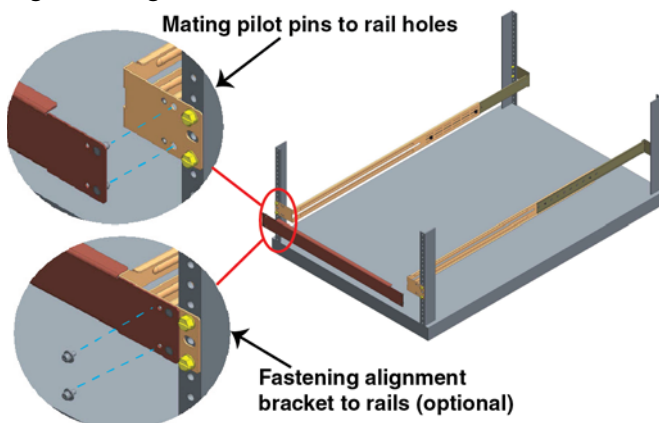
Loosely attach each rail assembly to the front and rear of the frame using two rack screws: one at the top pilot hole of the rail bracket and the other at the bottom pilot hole.

Figure 6: Rail assembly installation

6. Insert the alignment bracket.

Position the alignment bracket (410-7782-001) at the front of the frame by inserting the two pilot pins on each end of the alignment bracket into the rail holes on each front rail assembly.

Note: You can temporarily fasten the front alignment bracket to the rails using two (2) pan head machine screws (312-0330-M06) on each side to increase frame stability while installing the rear cable management bar.

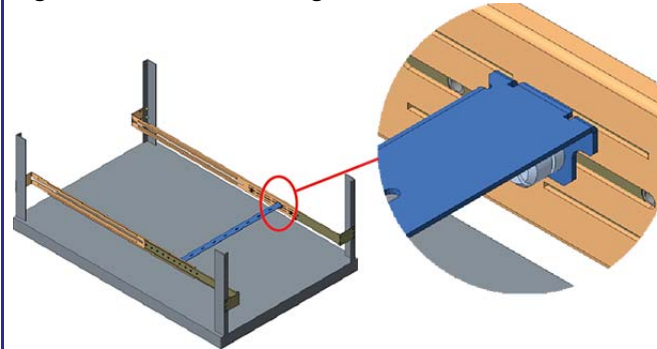
Figure 7: Alignment bracket insertion

7. Install the rear cable management bar.

With the cable management bar tab-side up in the frame, insert the tab at each end of the bar in a guide on the front rail. Tighten the thumb screw on each side of the bar to secure it to the rail assemblies. Do not over-tighten the thumb screws.

Note: The rear cable management bar helps maintain the rigidity of the rails during Waveserver Ai installation. Ciena recommends placing the bar far enough back in the frame so that there is adequate clearance to install

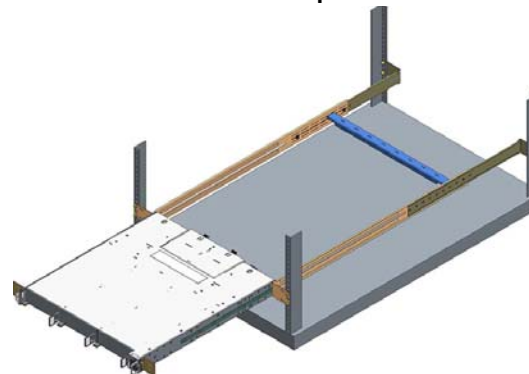
and/or remove modules, but close enough to the center to stabilize the rails.

Figure 8: Rear cable management bar installation

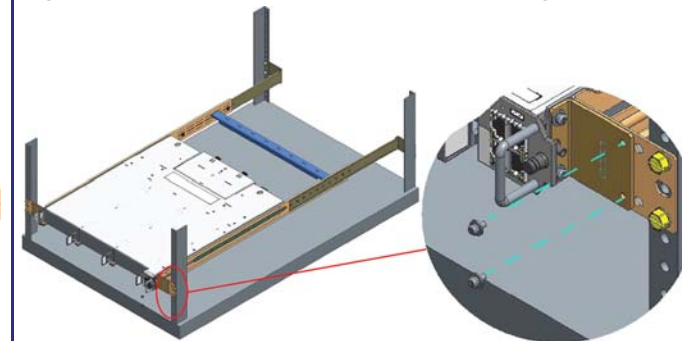
8. Firmly tighten the eight (8) frame mounting screws on the front and rear rail assemblies but do not fully torque, as some minor adjustments may be required when installing Waveserver Ai.

9. Remove the front alignment bracket. The rear cable management bar should be left in place for added frame stability.

10. Install the Waveserver Ai chassis in the equipment frame. Holding the chassis securely at the front of the frame, align the T-Bars on each side of the chassis with the pilot slot on each front rail assembly. Carefully slide the chassis into position in the frame until the frame-mounting bracket at each side of the chassis comes into contact with the bracket on the front rail.

Figure 9: Waveserver Ai chassis positioned for installation

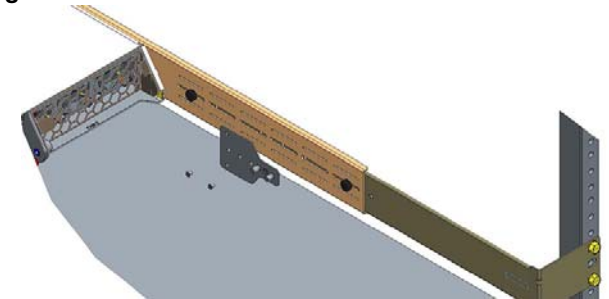
11. Secure each frame-mounting bracket to the frame using pan head machine screws (312-0330-M06) inserted in the two pilot holes of the front rail bracket on each side.

Figure 10: installation on front frame mounting bracket

Note: A chassis ground is required for all DC Waveserver Ai installations as well as AC Waveserver Ai installations requiring GR-1089-CORE compliance. For AC installations that do not require GR-1089-CORE compliance, the chassis ground is recommended but not required.

12. Install the ground bracket (410-6903-002).

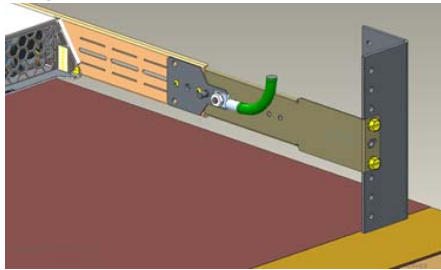
Install the ground bracket as far back as possible to maximize clearance for the installation and removal of the fan modules. Align the pilot holes on the ground bracket with the pilot holes on the slide rail assembly and fasten using wafer head machine screws (312-3730-CS1). Depending on the ground bracket positioning, either two or three of the three positions will be available. Use a minimum of two mounting positions.

Figure 11: Ground bracket installed

13. Install the chassis ground.

Note: Both single and two-hole ground lugs are supported. For DC Waveserver Ai installations, Ciena recommends using Molex 19164-0889 (ring lug with crimp terminal) to connect the DC power plant to the DC power module.

Crimp the ground wire into a ground lug. Fasten the lug to the ground bracket using Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m). Route the ground wire from the ground bracket and fasten to an equipment frame ground location.

Figure 12: DC ground wire installed**14. Install the power modules.**

Holding the module by its handle, carefully slide it into its slot at the rear of the Waveserver Ai chassis until you hear a click, which indicates that the holding clip has engaged.

Note: Waveserver Ai must be equipped with two power modules of the same model for 1:1 redundancy (Protection:Working). Under normal operating conditions, the combined power modules support the full Waveserver Ai power demand. When a power module fails or is removed from the chassis, the remaining power module assumes the full Waveserver Ai power demand without traffic interruption.

Figure 13: Power module installation**Warning**

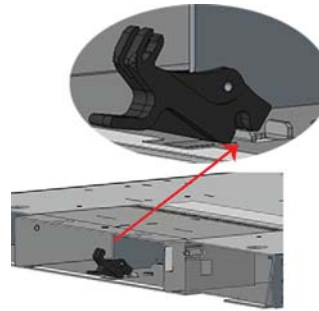
AC or DC power feeds to each power module must be rated to support the full power budget of an installed Waveserver Ai configuration. If one or both power feeds are underrated, the Waveserver Ai will shut down when a power module fails or is removed.

If multiple Waveserver Ais are powered with an external power distribution unit (PDU), each power input feed to the PDU must be rated for the aggregate capacity of all power modules powered by it.

15. Install the Waveserver Ai Control Subsystem.

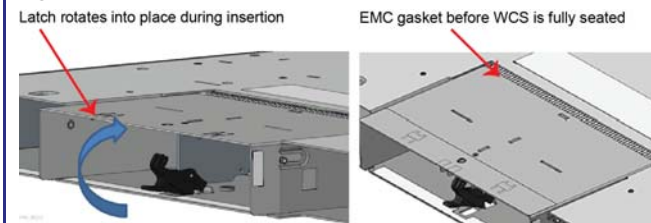
To install the WCS (186-1030-900), open the latch and slide the WCS into the chassis slot. Ensure that the latch pawl clears the lower flange as shown below.

Note: It is not necessary to use the latch during insertion. The latch is used to retain the WCS during operation and to disengage the WCS during removal.

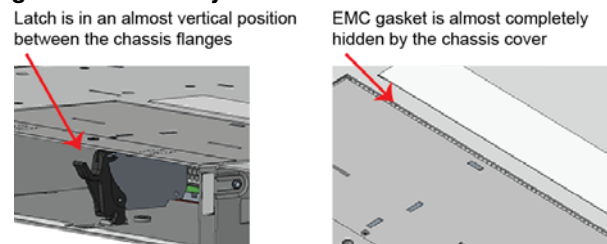
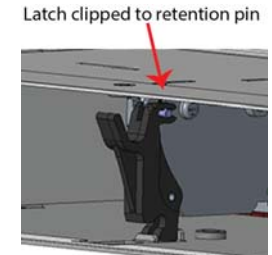
Figure 14: Installation of WCS in chassis slot- latch in open position

Slide the WCS into the slot, using even pressure along the top of the WCS cover, until it is fully seated. As the WCS is inserted into the slot, the latch contacts the insertion flange and starts to rotate into place.

Note: The EMC gasket contacts the chassis before the WCS is fully seated. Push the WCS past this contact point so that the EMC gasket is almost completely hidden by the chassis.

Figure 15: WCS installation**16. Verify that the WCS is fully seated in the Waveserver Ai chassis. The WCS is fully seated when:**

- the WCS EMC gasket is almost entirely hidden by the chassis.
- the retention latch is between the chassis flanges and in an almost vertical position.

Figure 16: WCS fully seated in the Waveserver Ai chassis**17. Completely rotate the latch until it clips onto the retention pin.****Figure 17: WCS latch fastened to retention pin****18. Install the fan modules.**

Holding the fan module by the handle, carefully slide it into its slot behind the WCS until you hear a click, which indicates that the holding clip has engaged.

Figure 18: Fan module installation**19. Install the modules.**

Install the modules at the front of the Waveserver Ai chassis, from left to right. Carefully slide the module into the slot until you hear a click, which indicates that the holding clip has engaged.

Figure 19: Modules installed**20. Install filler modules in empty slots.**

Carefully slide the filler module (186-1032-900) into the slot until you hear a click, which indicates that the holding clip has engaged.

21.

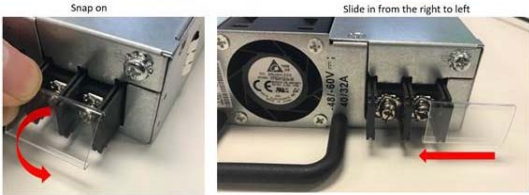
If you are installing	Then go to
a Waveserver Ai with DC power modules (screw terminals)	step 22
a Waveserver Ai with AC power modules (IEC connectors)	step 23

22. Prepare the DC power modules.

Using wire strippers, strip the DC power cables to 5/8-in. Crimp ring lugs onto the ends of the DC power wires. Attach the ring lugs to the “-” (-48Vdc) and “+” (Return) receptacles of the DC power module and fasten using a 1/4-in Phillips screwdriver to a torque value of 24 lb-in (2.7 N-m).

Slide or snap the protective plastic cover onto the terminal block.

Figure 20: Cover installation on terminal block (DC power modules)



Note 1: Ciena recommends using a minimum 10 AWG/6 mm² stranded copper cable with insulation rated for a minimum of 90°C. Solid cable is not supported.

Note 2: See the following table for the recommended wire gauge per maximum wire length from the Waveserver Ai power modules to the power distribution area.

Figure 21: DC power module



Table 1: Power cable sizes and maximum allowable cable distances from Waveserver Ai DC power module to the power distribution area:

Wire gauge	Maximum length from Waveserver Ai DC power module to power distribution area
Imperial	
AWG 8	8.3 m (27 ft)
AWG 10	5.2 m (17 ft)
AWG 12	3.3 m (11 ft)
Metric	
4 mm ²	4.0 m (13 ft)
6 mm ²	6.0 m (20 ft)
8 mm ²	7.9 m (26 ft)

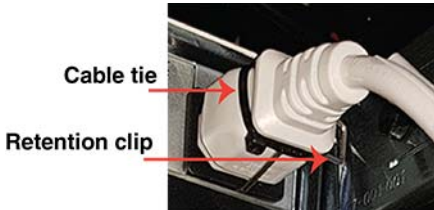
Note: This information is provided for reference only and does not necessarily consider any de-rating due to bundling of power cables, etc. The stated values are engineered to meet the ANSI T1.315 (generic industry standard) which states that the V-drop from the power distribution bay to the powered equipment must not exceed 1V. You must follow all local safety standards required by your company when determining the actual feeder size and maximum allowable distances.

23. Connect Waveserver Ai to power.

Route two AC or two DC power cables to the rear of the chassis and use electrical-insulation paper and binding to secure the cables to the rear cable management bar. Connect a cable to the receptacle at the each power module and then connect it to a power source.

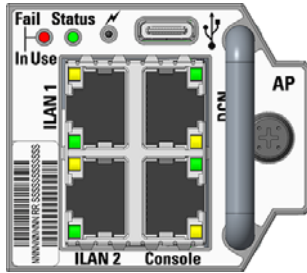
Note: AC and AC/240Vdc power modules are equipped with retention clips to prevent accidental disconnection of the cable. Since not all AC cables are compatible with the clips, perform a pull test after installing the cables. If the retention clip does not adequately secure the cable, use a cable tie to attach the cable to the clip.

Figure 22: AC power cable secured to retention clip



24. Observe the LEDs on the access panel as Waveserver Ai initializes. The red LED lights steadily for several seconds and turns off when the green Status LED lights.

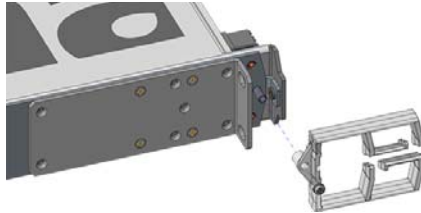
Figure 23: Waveserver Ai LED display



25. Install the fiber management clips.

Slide a fiber management clip onto the arm at each side of the Waveserver Ai chassis until it snaps into place. Secure the clip with pan head screw (312-0630-M08) using a hex driver.

Figure 24: Fiber management clip



26. Route the fibers.

Fibers can all be routed to the left side of the chassis, or the fibers from the left and middle modules can be routed left and the fibers from the right module routed right. If you are routing all of the fibers to the left, Ciena recommends that the fibers from the right-most module be routed first through the rear section of the clip. Fibers from the middle module should be routed through the lower front section and fibers from the left-most module should be routed through the upper front section.

Note: Other fiber routing configurations are also possible.

Figure 25: Fiber management and routing



Note: Ethernet and console port cables should be routed

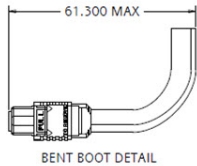
through the lower front section of the right fiber management clip (facing Waveserver Ai).

Figure 26: Routing and retention for Ethernet and console port cables



Note: When using the MPO fibers with a formable boot, make sure that you bend the boot at 90 degrees and secure the patch cord in the fiber management clip.

Figure 27: Formable boot bent at 90 degrees



27. Waveserver Ai is now ready to be commissioned. See "Commissioning Waveserver Ai" in *Waveserver Ai User Guide*, 323-4002-100.